

Research Notes on Part of speech tagging

Part of speech tagging

>> Section 1

Tag sets Schools commonly teach that there are 9 parts of speech in English: noun, verb, article, adjective, preposition, pronoun, adverb, conjunction, and interjection. However, there are clearly many more categories and sub-categories. For nouns, the plural, possessive, and singular forms can be distinguished. In many languages words are also marked for their "case" (role as subject, object, etc.), grammatical gender, and so on; while verbs are marked for tense, aspect, and other things. In some tagging systems, different inflections of the same root word will get different parts of speech, resulting in a large number of tags. For example, NN for singular common nouns, NNS for plural common nouns, NP for singular proper nouns (see the POS tags used in the Brown Corpus). Other tagging systems use a smaller number of tags and ignore fine differences or model them as features somewhat independent from part-of-speech. In part-of-speech tagging by computer, it is typical to distinguish from 50 to 150 separate parts of speech for English. Work on stochastic methods for tagging Koine Greek (DeRose 1990) has used over 1,000 parts of speech and found that about as many words were ambiguous in that language as in English. A morphosyntactic descriptor in the case of morphologically rich languages is commonly expressed using very short mnemonics, such as Ncmsan for Category = Noun, Type = common, Gender = masculine, Number = singular, Case = accusative, Animate = no. The most popular "tag set" for POS tagging for American English is probably the Penn tag set, developed in the Penn Treebank project. It is largely similar to the earlier Brown Corpus and LOB Corpus tag sets, though much smaller. In Europe, tag sets from the Eagles Guidelines see wide use and include versions for multiple languages. POS tagging work has been done in a variety of languages, and the set of POS tags used varies greatly with language. Tags usually are designed to include overt morphological distinctions, although this leads to inconsistencies such as case-marking for pronouns but not nouns in English, and much larger cross-language differences. The tag sets for heavily inflected languages such as Greek and Latin can be very large; tagging words in agglutinative languages such as Inuit languages may be virtually impossible. At the other extreme, Petrov et al. have proposed a "universal" tag set, with 12 categories (for example, no subtypes of nouns, verbs, punctuation, and so on). Whether a very small set of very broad tags or a much larger set of more precise ones is preferable, depends on the purpose at hand. Automatic tagging is easier on smaller tag-sets.

>> Section 2

Research on part-of-speech tagging has been closely tied to corpus linguistics. The first major corpus of English for computer analysis was the Brown Corpus developed at Brown University by Henry Kučera and W. Nelson Francis, in the mid-1960s. It consists of about 1,000,000 words of running English prose text, made up of 500 samples from randomly chosen publications. Each sample is 2,000 or more words (ending at the first sentence-end after 2,000 words, so that the corpus contains only complete sentences). The Brown Corpus was painstakingly "tagged" with part-of-speech markers over many years. A first approximation was done with a program by Greene and Rubin, which consisted of a huge handmade list of what categories could co-occur at all. For example, article then noun can occur, but article then verb (arguably) cannot. The program got about 70% correct. Its results were repeatedly reviewed and corrected by hand, and later users sent in errata so that by the late 70s the tagging was nearly perfect (allowing for some cases on which even human speakers might not agree). This corpus has been used for innumerable studies of word-frequency and of part-of-speech and inspired the development of similar "tagged" corpora in many other languages. Statistics derived by analyzing it formed the basis for most later part-of-speech tagging systems, such as CLAWS and VOLSUNGA. However, by this time (2005) it has been superseded by larger corpora such as the 100 million word British National Corpus, even though larger corpora are rarely so thoroughly curated. For some time, part-of-speech tagging was considered an inseparable part of natural language processing, because there are certain cases where the correct part of speech cannot be decided without understanding the semantics or even the pragmatics of the context. This is extremely expensive, especially because analyzing the higher levels is much harder when multiple part-of-speech possibilities must be considered for each word.

>> Section 3

The methods already discussed involve working from a pre-existing corpus to learn tag probabilities. It is, however, also possible to bootstrap using "unsupervised" tagging. Unsupervised tagging techniques use an untagged corpus for their training data and produce the tagset by induction. That is, they observe patterns in word use, and derive part-of-speech categories themselves. For example, statistics readily reveal that "the", "a", and "an" occur in similar contexts, while "eat" occurs in very different ones. With sufficient iteration, similarity classes of words emerge that are remarkably similar to those human linguists would expect; and the differences themselves sometimes suggest valuable new insights. These two categories can be further subdivided into rule-based, stochastic, and

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neural approaches.

>> Section 4

Works cited Charniak, Eugene. 1997. "Statistical Techniques for Natural Language Parsing". *AI Magazine* 18(4):33–44. Hans van Halteren, Jakub Zavrel, Walter Daelemans. 2001. Improving Accuracy in NLP Through Combination of Machine Learning Systems. *Computational Linguistics*. 27(2): 199–229. PDF DeRose, Steven J. 1990. "Stochastic Methods for Resolution of Grammatical Category Ambiguity in Inflected and Uninflected Languages." Ph.D. Dissertation. Providence, RI: Brown University Department of Cognitive and Linguistic Sciences. Electronic Edition available at [1] Archived 2018-08-19 at the Wayback Machine D.Q. Nguyen, D.Q. Nguyen, D.D. Pham and S.B. Pham (2016). "A Robust Transformation-Based Learning Approach Using Ripple Down Rules for Part-Of-Speech Tagging." *AI Communications*, vol. 29, no. 3, pages 409–422. [.pdf]

>> Section 5

Other taggers and methods Some current major algorithms for part-of-speech tagging include the Viterbi algorithm, Brill tagger, Constraint Grammar, and the Baum-Welch algorithm (also known as the forward-backward algorithm). Hidden Markov model and visible Markov model taggers can both be implemented using the Viterbi algorithm. The rule-based Brill tagger is unusual in that it learns a set of rule patterns, and then applies those patterns rather than optimizing a statistical quantity. Many machine learning methods have also been applied to the problem of POS tagging. Methods such as SVM, maximum entropy classifier, perceptron, and nearest-neighbor have all been tried, and most can achieve accuracy above 95%. A direct comparison of several methods is reported (with references) at the ACL Wiki. This comparison uses the Penn tag set on some of the Penn Treebank data, so the results are directly comparable. However, many significant taggers are not included (perhaps because of the labor involved in reconfiguring them for this particular dataset). Thus, it should not be assumed that the results reported here are the best that can be achieved with a given approach; nor even the best that have been achieved with a given approach. In 2014, a paper reported using the structure regularization method for part-of-speech tagging, achieving 97.36% on a standard benchmark dataset.